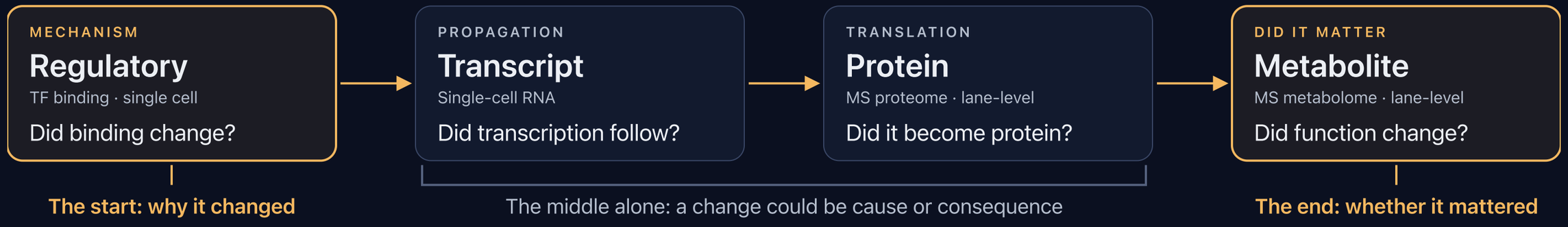
Why all four layers? Why is three not enough?

Three layers show that something changed. Only the two ends say why, and whether it mattered.





Sources | Baik 2019 10.1016/j.cmet.2019.06.005 · Liu, Beyer & Aebersold 2016 (transcript does not predict protein) 10.1016/j.cell.2016.03.014

Is this really single-cell?

Regulatory and transcript: one cell at a time. Protein and metabolite: lane-level, and we say so.

SINGLE CELL · ONE CELL PER CAPSULE



Each cell read on its own: its own binding, its own transcriptome

LANE-LEVEL · BULK



Pooled over the lane's cells

THIS ASSAY	Parallel fields of the same lanes: one population, one exposure history. The layers are comparable.
SEPARATE CULTURES	One culture per layer: different batch, different history. Differences between layers are confounded.

Sources | Single-cell layers: TIP-seq, Bartlett 2021 10.1083/jcb.202103078 · Lane-level MS layers and capsule barcoding: project design, unpublished. One population and one history, not one cell.

Once the cells are pooled, how do you know which chamber each one came from?

Label every chamber before harvest. Clustering finds the cell type, never the condition.

1 **Label each chamber before harvest** — PREFERRED

A lipid-anchored oligo on the cells, MULTI-seq style.
It travels with the cell, as Perturb-seq's guide does.

2 **Barcode each chamber at encapsulation**

A barcode assigned per chamber when the cells are packaged.
The alternative if the label cannot reach the library.

3 **Cluster the transcriptomes**

Gives CELL TYPE, not CONDITION.
Cannot replace 1 or 2.

POOLED, THEN CLUSTERED



Neurons

ten chambers, ten conditions,
one cluster. Which came from where?



Microglia



Astrocytes

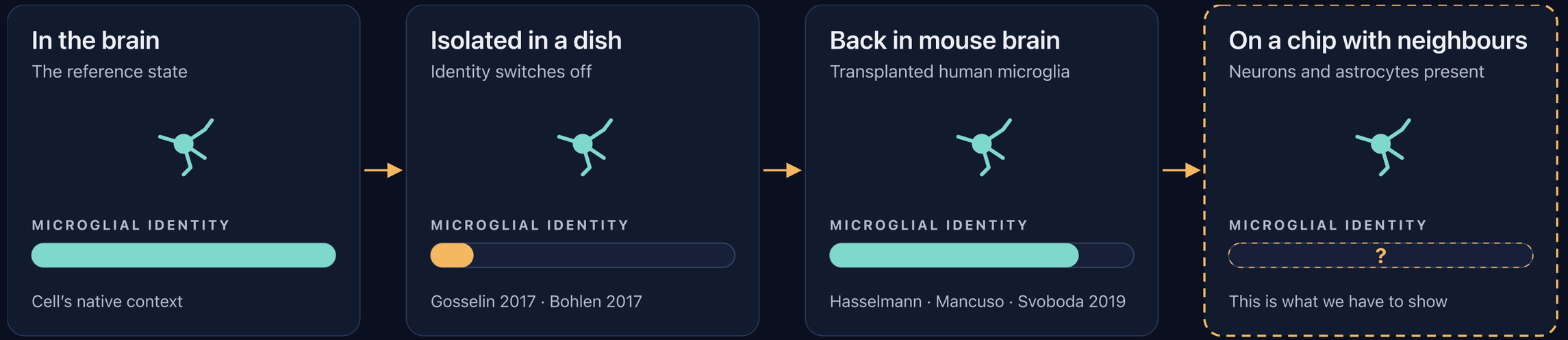
NOT YET VERIFIED that the label reaches our library

Sources | MULTI-seq, McGinnis 2019 10.1038/s41592-019-0433-8 · Perturb-seq, Dixit 2016 10.1016/j.cell.2016.11.038 · Whether a label reaches our library is unverified; project design, unpublished.

Will iPSC microglia be real microglia on a chip?

WE DO NOT KNOW YET

On a chip, not yet shown. Neighbours restore microglia in mouse brain; a chip has to prove it.



WHY THE PLATFORM EXISTS Step two is the problem. Neighbours are the proposed fix.

Bars are a schematic, not measured values. The chip result is something to demonstrate, not a claim.

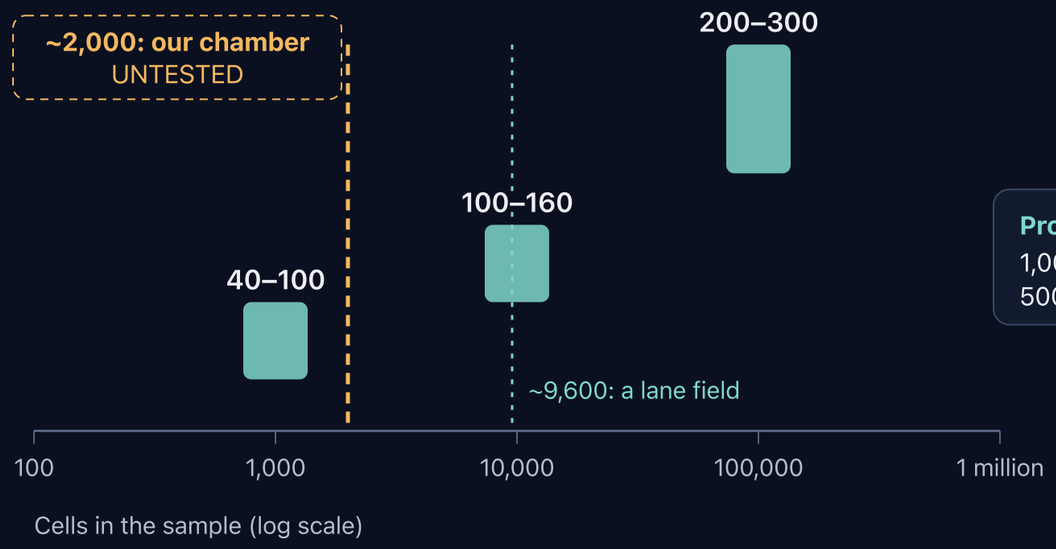
Sources | Gosselin 2017 10.1126/science.aal3222 · Bohlen 2017 10.1016/j.neuron.2017.04.043 · Hasselmann 2019 10.1016/j.neuron.2019.07.002 · Mancuso 2019 10.1038/s41593-019-0525-x · Svoboda 2019 10.1073/pnas.1913541116

Can you get a proteome and a metabolome from a few thousand cells?

WE DO NOT KNOW YET

Not yet tested at ~2,000 cells. The literature says proteome yes, metabolome partly, and never ATP or ADP.

METABOLITES IDENTIFIED vs CELLS IN THE SAMPLE (LITERATURE)



Proteome
1,000 cells give
500–1,000 proteins

STAY USABLE AFTER A FINITE RINSE

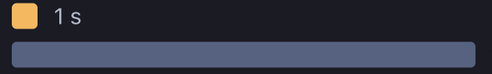
TCA intermediates
Amino acids · lipids

INVALID: WE GIVE THESE UP

ATP · ADP · energy charge

and glycolytic intermediates

Turnover is about 1 s. Our shortest rinse is about 18 s.



Sources | Proteome: Mueller 2020 10.15252/msb.20199111 · Rinse and leakage: Lorenz 2011 10.1021/ac103313x · Kapoore 2017 10.1039/C7AN00207F · Metabolite counts per cell number, lane size and rinse time: project literature summary and design records, unpublished; the ~2,000-cell input is untested.

Why three cell types? Why not an organoid?

We give up self-organisation to control who touches whom, and when. Three types is the proof of concept.





	ORGANOID	CHAMBERS (THIS PLATFORM)
Self-organisation	Yes, emerges on its own	**Given up**
Who touches whom	Whatever develops	**Set by us**
When they meet	Fixed by development	**Set by us, on command**
Single-type baselines	Not available	**One chamber per type**

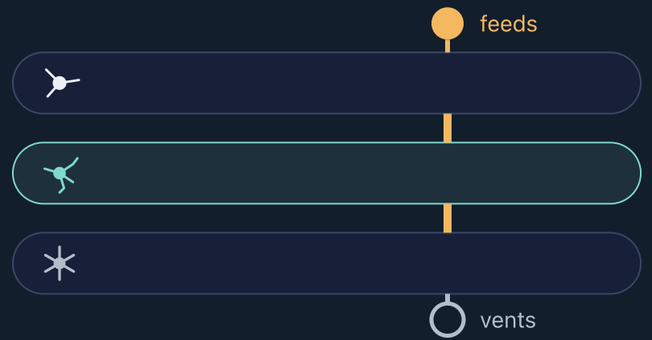
Next: 3 types now, 10 compartments and beyond; each added compartment multiplies the combinations.

Sources | Microglia change organoid neural cells: Popova 2021 `10.1016/j.stem.2021.08.015` · Three types is the lab's use case; the path to ten compartments is project design, unpublished.

How do you keep cultures apart for weeks, then connect them on command?

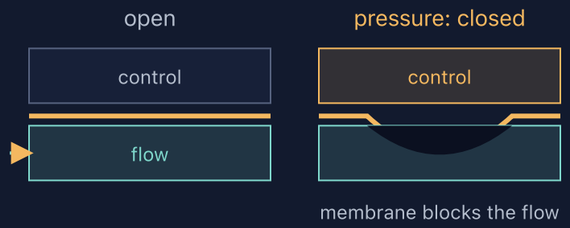
Today: separate lanes, routed by which port feeds and which vents. No valves. Valves are the full-spec route.

THE DEMO · SINGLE LAYER, NO VALVES



Route by which port feeds and which vents

FULL SPEC · PNEUMATIC VALVE



Two channels, a flexible membrane; pressure closes the flow channel

Needs a second layer plus a membrane

FULL SPEC · MICROCHANNEL BARRIER



Contact without mixing the cells

Needs photolithography

BUILT FOR THE DEMO Column one only. Columns two and three are on a different fabrication route.

OPEN TEST Microglia are motile; a barrier that stops neurons may not stop them.

Sources | Pneumatic valve: Unger 2000 10.1126/science.288.5463.113 · Thorsen 2002 10.1126/science.1076996 · Microchannel barrier: Taylor 2005 10.1038/nmeth777 · Demo layout: project design, unpublished.

Mature neurons will not lift off the plate. How do you harvest them?

TWO SPLIT DAYS UNRESOLVED

You don’t. Split at the last detachable precursor. Microglia: about day 50–52. Neuron and astrocyte days are open.



Sources | Microglia route: McQuade 2018 10.1186/s13024-018-0297-x · Washer 2022 10.1038/s41598-022-23477-2 · Neurons: Shi 2012 10.1038/nprot.2012.116 · Astrocytes: Schuurmans 2026 10.1002/glia.70202 · Day figures are project schedule, unpublished; the microglia day is ±2.

What would you run first, and what would you add to the medium?

First: who is present, in what ratio, and when each joins. Then dose and timing of soluble-factor classes.

1 · WHO IS PRESENT
7 combinations



single-type chambers are the baselines | pairs | all three

2 · RATIO
counted at plating



1 : 1 | 1 : 3 | 3 : 1

3 · WHEN EACH JOINS



together at plating | second type joins later

A then B is not B then A

4 · THEN ADD
soluble-factor classes



dose →, timing ↓

Dose and timing of each class, on top of the cell-type matrix

5 · AND



Genetic backgrounds, compared in one setting

Experimental design level only. Chambers of one type are the baselines for every comparison.

Sources | Experimental design of the platform: project design, unpublished. No recipes or concentrations are given here.

What is actually built?

Six chips printed, dye only. A pipeline that works. No biology has run.

WORKING

Six chips printed
Dye-only bench demo.
Single layer, valveless.

Protocol-to-chip pipeline
Ten steps, protocol in,
print files out.

Flow simulation
Already catches real faults
before printing.

DESIGNED, NOT BUILT

Full-scale design
Simulated, not built.

Valves and barriers
Full-spec options on a
different fabrication route.

Capsule barcoding stage
Designed, protocol written,
not printed or run.

NOT YET RUN

Any biology on a chip
Dye only so far.

MS from a ~2,000-cell chamber
Input needed is untested.

Condition label to library
Whether it reaches our
library is unverified.

UNPUBLISHED

Capsule barcoding combined with TIP-seq

Each half is published:

TIP-seq
Bartlett 2021
+
Capsules + barcoding

The combination is not.

Nobody has published it.

Barcoding half is a preprint.

Sources | Project design, simulation and bench records, unpublished. TIP-seq 10.1083/jcb.202103078 · Capsules, Leonaviciene & Mazutis 2023 10.1093/nar/gkac918 · AmpliDrop, Garcia-Bassets 2024, preprint 10.1101/2024.08.13.607508